

Kneo Agent Platform

Tutorial

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Quickstart, custom-tool tutorial, PostgreSQL deployment tutorial, and the human-in-the-loop walkthrough — one printable reading path.

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Quickstart

Source: <docs/user/quickstart.md>

A guided walkthrough that takes you from a fresh checkout of Kneo Agent Platform (`kneo-serv`) to a running agent — locally and through the service — in about 15 minutes. By the end you'll have validated a spec, run an agent, exercised a human-in-the-loop pause, and called the same workflow over HTTP.

This expands on the README quickstart; pick either entry point.

Prerequisites

- Python 3.12+ (`python --version`).
- An OpenAI-compatible API key for the demo spec (`OPENAI_API_KEY`). Other providers work, but the example uses OpenAI.
- Optional: Docker, if you want to follow the service path with PostgreSQL.

1. Install

Pick the path that matches what you're doing.

Trying it out / using it as an application — install from PyPI (available from `kneo-serv 0.2.0`):

```
python -m venv .venv && source .venv/bin/activate
python -m pip install kneo-serv
```

Contributing or hacking on the source — clone and install in editable mode with the dev extra:

```
git clone git@github.com:kneo-agent/kneo-serv.git
cd kneo-serv
python -m venv .venv && source .venv/bin/activate
python -m pip install -e ".[dev]"
```

Either path exposes the `kneo` command. Verify:

```
kneo --help
```

If you don't have an OpenAI key handy, swap to the dummy provider used by the smoke test (see [§ 5](#) below).

2. Validate and compile a spec

The repo ships a runnable research-agent spec.

```
kneo spec validate examples/research_agent.yaml
kneo spec compile examples/research_agent.yaml
```

`validate` runs schema + semantic checks and prints diagnostics; `compile` goes further and constructs the agent and workflow objects without running them. Both should exit 0.

If validation fails, the diagnostic message includes a path into the spec (`agent.tools[0]` , etc.) and a short reason. The most common causes are listed in [troubleshooting.md § 6](#).

3. Run locally

```
export OPENAI_API_KEY=sk-...
kneo run examples/research_agent.yaml \
  --input "Summarize Nvidia's AI strategy in three bullet points" \
  --target workflow
```

Local runs persist state at `.kneo/kneo_runs.sqlite` and continuations under `.kneo/continuations` . Inspect them:

```
kneo runs get <run_id>
kneo runs trace <run_id>
kneo runs checkpoints <run_id>
```

The run output lands in `RunResult.output_text` ; the CLI also prints it on stdout by default. Use `--json` to get the full structured response.

4. Try a human-in-the-loop workflow

```
kneo run examples/human_approval_workflow.yaml \
  --input "draft the launch announcement" \
  --target workflow --json
```

The JSON response includes a top-level `continuation_id` ; the `request_id` is nested under `metadata.pending_human_request` . Resume with an approval:

```
kneo human resume <continuation_id> --request-id <request_id> --approve
```

For a deeper end-to-end walkthrough, see [human_in_the_loop.md](#).

5. No real provider: the dummy path

The smoke spec uses the in-process `dummy` provider so it runs without real credentials.

```
kneo run examples/smoke_human_workflow.yaml \
  --input "smoke" --target workflow --json
```

This is the same flow the deployment smoke exercises; see [deployment_smoke.md](#).

6. Run as a service

In one terminal, start the API:

```
export KNEO_SERV_AUTH_ENABLED=true
export KNEO_SERV_API_KEYS='operator:operator-token:operator'
kneo service serve --host 127.0.0.1 --port 8000
```

In another, hit the API:

```
export KNEO_SERV_API_KEY=operator-token
curl -sf http://127.0.0.1:8000/livez
curl -sf http://127.0.0.1:8000/readyz | jq
```

Submit a run through the CLI's service mode:

```
kneo run examples/smoke_human_workflow.yaml \
  --service-url http://127.0.0.1:8000 \
  --input "hello" --target workflow --json
```

Or directly with curl:

```
curl -sf -X POST http://127.0.0.1:8000/v1/runs \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" \
  -H 'Content-Type: application/json' \
  -d '{
    "input": "hello",
    "spec_path": "examples/smoke_human_workflow.yaml",
    "target": "workflow"
  }'
```

For the full HTTP contract, see [service_api.md](#). For deployment topology (Docker, Compose, PostgreSQL), see [deployment.md](#).

7. Use a profile for repeated calls

CLI profiles save `--service-url` and `--api-key` so you don't type them on every command.

```
kneo config profile set local \
  --service-url http://127.0.0.1:8000 \
  --api-key operator-token
kneo config profile use local

kneo run examples/smoke_human_workflow.yaml \
  --input "hello" --target workflow --profile local
kneo human list --profile local
```

Profiles live at `~/.kneo_serv/profiles.json` (or `KNEO_SERV_PROFILES_PATH`) with owner-only file permissions. Stored tokens are never printed by the CLI.

Where to go next

You want to...	Go to
Write your own spec	examples.md , project_config.md
See every CLI subcommand and flag	cli_reference.md
Read the full HTTP contract	service_api.md
Deploy with Docker / Compose / PostgreSQL	deployment.md
Tune env vars	environment.md
Diagnose a failure	troubleshooting.md
Add a custom tool, runtime, or store	extending.md
Understand the architecture	design.md
Look up jargon	glossary.md

Tutorial: writing a custom tool end-to-end

Source: docs/user/tutorial_custom_tool.md

Build a simple custom tool, register it with `kneo-serv`, expose it in a YAML spec, run an agent that calls it, and verify how the call shows up in audit and trace events.

This tutorial walks the path from "I have a Python function" to "an agent uses it in production" so you can see every layer the call passes through.

For the recipe-style summary, see [extending.md § 1](#). For the public-API surface, see [implementation_map.md § tools/](#).

What we're building

A `lookup_user` tool that takes a user id and returns the user's email. We'll start with a function, register it as a tool, and run a small agent that uses it.

1 · Write the Python function

Create `examples/my_tools.py` (or anywhere on the import path):

```
# examples/my_tools.py
from typing import Any

USERS = {
    "u-001": "alice@example.com",
    "u-002": "bob@example.com",
    "u-003": "carol@example.com",
}

def lookup_user(args: dict[str, Any]) -> str:
    """Return the email for a user id, or 'unknown' if not present."""
    user_id = args.get("user_id")
    if not user_id:
        return "unknown: user_id is required"
    return USERS.get(user_id, f"unknown: {user_id}")
```

Two things to notice:

- The handler signature is `Callable[[dict[str, Any]], str]`. Args arrive as a dict; the return value must be a string. The framework wraps it as a `ToolResult`.

- The function should be deterministic and side-effect-free where possible. If it does I/O, plan for retries and timeouts (see [environment.md § Runtime Reliability](#)).

2 · Register the tool with ToolRegistry

Tools live in a `ToolRegistry`. The default service registry is built in [service/factory.py](#); to add a custom tool you need a registry that includes it. The cleanest approach is to construct a custom registry and pass it into the platform manager.

Create `examples/my_factory.py`:

```
# examples/my_factory.py
from kneo_serv.platform import PlatformManager
from kneo_serv.spec import SpecCompiler
from kneo_serv.service.factory import (
    create_runtime_registry,
    create_tool_registry,
    create_persistence_stores,
)
from kneo_serv.tools import ToolDefinition

from examples.my_tools import lookup_user

def build_platform() -> PlatformManager:
    runtime_registry = create_runtime_registry()
    tool_registry = create_tool_registry(include_examples=True)
    tool_registry.register(
        ToolDefinition(
            name="lookup_user",
            description="Return the email for a user id.",
            parameters={
                "type": "object",
                "properties": {"user_id": {"type": "string"}},
                "required": ["user_id"],
            },
        ),
        lookup_user,
    )

    compiler = SpecCompiler(
        runtime_registry=runtime_registry,
        tool_registry=tool_registry,
    )
    run_state, continuation = create_persistence_stores()
    manager = PlatformManager(
        compiler=compiler,
```

```

        run_state_store=run_state,
        continuation_store=continuation,
    )
    manager.start_worker()
    return manager

```

The `parameters` schema is JSON Schema. The agent's tool-call planner sees this — be explicit about types, required fields, and enums. The guarded tool registry applies any policy from the spec on top of this definition, so you don't need to enforce auth here.

3 · Reference the tool from a spec

Create `examples/lookup_agent.yaml`:

```

version: v1

agent:
  name: directory-agent
  system_prompt: |
    You answer questions by looking up users with the lookup_user tool.
    Return concise, factual answers.
  model:
    provider: openai
    name: gpt-4o-mini
  strategy:
    type: react
    max_iterations: 4
  runtime_preferences:
    preferred_mode: bridge
    allowed_modes: [bridge]
  tools:
    include: [lookup_user]

workflow:
  type: sequential
  name: lookup-pipeline
  steps:
    - id: answer
      kind: agent
      ref: directory-agent

```

The `tools.include` list names tools the agent is allowed to call. Names must match what we registered in step 2. Validate the spec:

```
kneo spec validate examples/lookup_agent.yaml
```

If you see `E_T00L_REF`, the spec is referencing a name that isn't in the registry. Re-check step 2 — the registry must include `lookup_user` before `SpecCompiler` runs.

4 · Run the agent

When using a custom factory, drive the agent through Python rather than the bare `kneo run` command — the CLI's default platform manager does not include your custom tool.

```
# examples/run_lookup.py
from examples.my_factory import build_platform

def main() -> None:
    manager = build_platform()
    result = manager.run_from_spec(
        input_text="What's the email for u-002?",
        spec_path="examples/lookup_agent.yaml",
        target="workflow",
    )
    print(f"run_id={result.run_id}")
    print(result.output_text)

if __name__ == "__main__":
    main()
```

Run it:

```
export OPENAI_API_KEY=sk-...
python -m examples.run_lookup
```

Expected output (model wording will vary):

```
The email for u-002 is bob@example.com.
```

If the agent didn't call the tool, increase `max_iterations` or refine the system prompt to instruct tool use.

5 · Inspect the call in trace and audit

The tool call shows up in the run trace. Capture the `run_id` the script prints and pass it to `kneo runs trace`:

```
RUN_ID=$(python -m examples.run_lookup | sed -n 's/^run_id=//p')
kneo runs trace "$RUN_ID"
```

Look for `tool_call_started` and `tool_call_completed` events with `tool_name:` `"lookup_user"`. Tool arguments and results are *not* in the trace by default — they're redacted at write time. To capture them in OpenTelemetry spans for a specific deployment, opt in with `KNEO_SERV_OTEL_RECORD_ARGUMENTS=true` and `KNEO_SERV_OTEL_RECORD_RESULTS=true` (only after a data-classification review; see [environment.md § Observability](#)).

The audit log records the run (not individual tool calls) at `run.created` / `run.cancelled` / `run.continued`. Tool arguments are not persisted in audit events under any flag — see [production_readiness_review.md § Audit Payload Review](#).

6 · Lock down the tool with policy

For production, restrict what the agent can do with your tool. Add to the spec under `agent`:

```
agent:
  # ... existing fields ...
  policies:
    tool:
      allow:
        - lookup_user
      deny: []
      network: false
      filesystem: false
      shell: false
```

The guarded registry blocks every tool not in `allow`, and diagnostics check that registered tools don't claim capabilities (network, filesystem, shell) inconsistent with the policy.

Validate and re-run; the policy is enforced at call time. A blocked call surfaces as a `tool_policy_denied` trace event and never reaches your handler.

7 · Ship the tool to the service

Two patterns:

- **In-process custom factory.** Replace `service.factory` with your own `create_default_platform_manager()` that registers the tool. Run `kneo service serve` from a deployment that imports your factory. This is the common shape; the Docker image is a thin wrapper around `service.app:create_app(configure_default_manager=True)`, and you can override the manager in your own entrypoint.

- **Custom server.** Construct the FastAPI app yourself with `create_app(configure_default_manager=False)` and call `kneo_serv.service.dependencies.set_platform_manager(your_manager)` before serving. Useful when you want to register multiple tools, or combine custom auth + custom tools.

```
# examples/serve_lookup.py
import uvicorn
from kneo_serv.service.app import create_app
from kneo_serv.service.dependencies import set_platform_manager

from examples.my_factory import build_platform

if __name__ == "__main__":
    set_platform_manager(build_platform())
    uvicorn.run(create_app(), host="127.0.0.1", port=8000)
```

```
python -m examples.serve_lookup
curl -sf http://127.0.0.1:8000/livez
```

Now the same `lookup_user` tool is reachable from any spec that references it, including service-backed CLI calls and HTTP `POST /v1/runs`.

Common pitfalls

Symptom	Likely cause
<code>E_TOOL_REF</code> on <code>kneo spec validate</code>	Custom tool isn't in the registry the CLI is using; use the Python entrypoint from § 4.
<code>ValueError: Tool 'X' has no implementation</code>	<code>ToolDefinition.name</code> doesn't match the spec's <code>tools.include</code> entry.
Agent never calls the tool	System prompt doesn't mention it, or <code>max_iterations</code> is too low.
Tool calls succeed but content is missing in trace	Expected — tool args/results are redacted by default.
<code>403 Forbidden</code> when calling through service	API key is missing the <code>runs:write</code> scope; see troubleshooting.md § 4.2 .

Next

- More examples and orchestration patterns: [examples.md](#) .
- Custom MCP servers (similar pattern, externally hosted tools): [extending.md § 2](#).
- Custom middleware around tool calls (rate-limiting, logging): [extending.md § 3](#).

Tutorial: deploying with PostgreSQL from zero

Source: docs/user/tutorial_postgres_deployment.md

End-to-end deployment of `kneo-serv` against PostgreSQL using the bundled Docker Compose stack: rendering env files, starting the service, verifying readiness, and running smoke tests. Budget about 30 minutes from a fresh checkout to a running deployment.

For the reference on deployment shapes and persistence selection, see <deployment.md>. For environment-variable semantics, see <environment.md>.

Prerequisites

- Docker 24+ and `docker compose`.
- `git`, `curl`, `jq`, and a shell that supports `$( )` substitution.
- `python3 ≥ 3.12` for running the deployment-smoke script.
- Network access to pull the `postgres:16` and Python base images.

This tutorial uses 127.0.0.1; for a real deployment, substitute your host or load-balancer URL throughout.

1 · Clone and prepare the env file

```
git clone git@github.com:kneo-agent/kneo-serv.git
cd kneo-serv
cp deploy/production.env.example deploy/production.env
chmod 600 deploy/production.env
```

`deploy/production.env` is gitignored — it'll hold your real secrets. Edit it now and replace every `replace-...` placeholder. The minimum set you must change before binding to a network:

```
# deploy/production.env

# Auth – replace each token with a high-entropy value.
KNEO_SERV_AUTH_ENABLED=true
KNEO_SERV_API_KEYS=operator:OP_TOKEN:operator;reviewer:REV_TOKEN:reviewer;viewer:VIEW_TOKEN:viewer
KNEO_SERV_ADMIN_API_KEY=ADMIN_TOKEN
KNEO_SERV_SPEC_SIGNING_KEY=SPEC_SIGNING_HMAC_KEY

# PostgreSQL – match the password to the one you'll set below.
POSTGRES_DB=kneo_serv
```

```
POSTGRES_USER=kneo_serv
POSTGRES_PASSWORD=DB_PASSWORD
```

Generate strong tokens:

```
# 32-byte hex tokens
for name in OP_TOKEN REV_TOKEN VIEW_TOKEN ADMIN_TOKEN SPEC_SIGNING_HMAC_KEY DB_PASSWORD; do
    printf '%s=%s\n' "$name" "$(openssl rand -hex 32)"
done
```

Paste the generated values into `deploy/production.env`. Do not commit this file.

2 · Validate the env file

There's a validator that catches common mistakes (placeholder tokens, incomplete scoped roles, missing DSN, telemetry payload capture left on by accident):

```
python scripts/validate_staging_env.py --env-file deploy/production.env
```

Address any errors before continuing. Common findings:

- `replace-...` strings still present.
- Scoped role list missing `operator`, `reviewer`, or `viewer`.
- `KNEO_SERV_OTEL_RECORD_ARGUMENTS=true` without an explicit data-classification override.

3 · Start the Compose stack

The stack runs the API plus PostgreSQL with a persistent volume.

```
docker compose --env-file deploy/production.env up --build -d
```

`--build` rebuilds the API image so any local edits land. `-d` runs detached. To watch logs:

```
docker compose logs -f api
```

You should see the API come up after PostgreSQL passes its healthcheck (about 10–20 seconds on a cold start). The API logs include a line per migration applied at first startup.

4 · Verify readiness


```
export BASE=http://127.0.0.1:8000
curl -sf "$BASE/livez"
curl -sf "$BASE/readyz" | jq
```

`/livez` returns `{"ok": true, "metadata": {"status": "alive"}, ...}` as soon as the process accepts connections. `/readyz` returns 200 only after every dependency check passes:

```
{
  "ok": true,
  "metadata": {
    "ready": true,
    "manager": "PlatformManager",
    "checks": {
      "run_state_store": {"name": "run_state_store", "ok": true},
      "continuation_store": {"name": "continuation_store", "ok": true},
      "queue": {"name": "queue", "ok": true},
      "runtime_registry": {"name": "runtime_registry", "ok": true, "count": 2, "names":
["bridge", "native"]},
      "tool_registry": {"name": "tool_registry", "ok": true, "count": 3, "names": ["summarize",
"web_search", "webpage_reader"]},
      "providers": {"name": "providers", "ok": true},
      "mcp": {"name": "mcp", "ok": true}
    }
  }
}
```

If you get a 503, the body identifies which check failed; see [troubleshooting.md § 1.2](#).

5 · Run the deployment smoke

The smoke script exercises the full path: auth, spec validation, run creation, human resume, audit listing, credential inventory, and policy update.

```
export OP_TOKEN=<your-operator-token>
export REV_TOKEN=<your-reviewer-token>
export VIEW_TOKEN=<your-viewer-token>

python scripts/deployment_smoke.py \
  --base-url "$BASE" \
  --operator-api-key "$OP_TOKEN" \
  --reviewer-api-key "$REV_TOKEN" \
  --viewer-api-key "$VIEW_TOKEN"
```

A clean run prints each step with a `PASS` and exits 0. If any step fails, the script identifies the failing endpoint and HTTP status. See [deployment_smoke.md](#) for the full step list and what each step

verifies.

6 · Submit a real run

```
curl -sf -X POST "$BASE/v1/runs" \
  -H "Authorization: Bearer $OP_TOKEN" \
  -H 'Content-Type: application/json' \
  -d '{
    "input": "smoke",
    "spec_path": "examples/smoke_human_workflow.yaml",
    "target": "workflow"
  }' | jq
```

This spec uses the in-process `dummy` provider so it runs without provider credentials. You should see a `blocked` response with a `continuation_id` (the workflow has a human step). Resume it:

```
curl -sf -X POST "$BASE/v1/human-tasks/cont_.../resume" \
  -H "Authorization: Bearer $REV_TOKEN" \
  -H 'Content-Type: application/json' \
  -d '{"request_id": "req_...", "decision": "approved"}' | jq
```

For the full HITL flow, see [human_in_the_loop.md](#).

7 · Verify persistence survives restart

Confirm PostgreSQL volume persistence:

```
docker compose --env-file deploy/production.env restart api
sleep 5
curl -sf "$BASE/v1/runs?limit=5" \
  -H "Authorization: Bearer $OP_TOKEN" | jq '.runs[].run_id'
```

You should see the run id from step 6 in the list, even after the API container restarts. The data lives in the `postgres-data` named volume, not the container layer.

8 · Capacity tuning knobs

For a real production deployment, revisit these env vars in `deploy/production.env` after you have load profile data:

Variable	Default	Tune when
KNEO_SERV_PROVIDER_TIMEOUT_SECONDS	unset (suggest 120)	Provider tail latency exceeds default.
KNEO_SERV_PROVIDER_RETRIES	0 (suggest 2)	Provider has documented transient error rate.
KNEO_SERV_MAX_BODY_BYTES	1 MiB	You receive larger inline specs or override payloads.
KNEO_SERV_MAX_INPUT_CHARS	20000	Run inputs are larger than the default.
KNEO_SERV_RETENTION_RUNS_DAYS	unset	Storage growth requires capping run history.
KNEO_SERV_CHECKPOINT_COMPRESS_BYTES	64 KiB	Many large checkpoints; reduce to compress more.

Canonical defaults and tuning guidance: [environment.md](#) and [performance.md](#).

9 · Backup the database

A periodic seeded backup/restore drill is recommended operator practice — run one on a regular cadence and again immediately before any major upgrade (see [backup_and_recovery.md](#)). The shape for production:

```
# Backup
docker compose --env-file deploy/production.env exec db \
  pg_dump -U "$POSTGRES_USER" "$POSTGRES_DB" \
  | gzip > "kneo_serv-$(date +%Y%m%d-%H%M).sql.gz"

# Restore (DESTRUCTIVE – wipes current data)
gunzip -c kneo_serv-YYYYmmDD-HHMM.sql.gz \
  | docker compose --env-file deploy/production.env exec -T db \
    psql -U "$POSTGRES_USER" "$POSTGRES_DB"
```

The full drill — including verifying that runs, checkpoints, audit events, and policy metadata survive a restore — is in [release_checklist.md](#).

10 · Tear down

To stop the stack but keep data:

```
docker compose --env-file deploy/production.env down
```

To stop the stack *and* delete the PostgreSQL volume (destroys runs, checkpoints, audit events, continuations):

```
docker compose --env-file deploy/production.env down --volumes
```

Use the volume-deleting form for clean re-tests; never run it against a production deployment without a verified backup.

Common failure modes

Symptom	See
API container restarts with <code>KNEO service auth</code> is enabled but no API keys are configured	troubleshooting.md § 1.1
<code>/readyz 503</code> with <code>run_state_store</code> not ok	troubleshooting.md § 2.2
Service writes to SQLite even with DSN set	troubleshooting.md § 2.1
Smoke script fails on policy write	API key probably missing <code>policies:write</code> ; see troubleshooting.md § 4.2

Where to go next

- [staging_release_runbook.md](#) — promotion path beyond a single host.
- [deployment.md](#) — reference for deployment shapes, including running without Compose.
- [environment.md](#) — every env var.
- [tutorial_custom_tool.md](#) — extend this deployment with custom tools.

Human-in-the-loop walkthrough

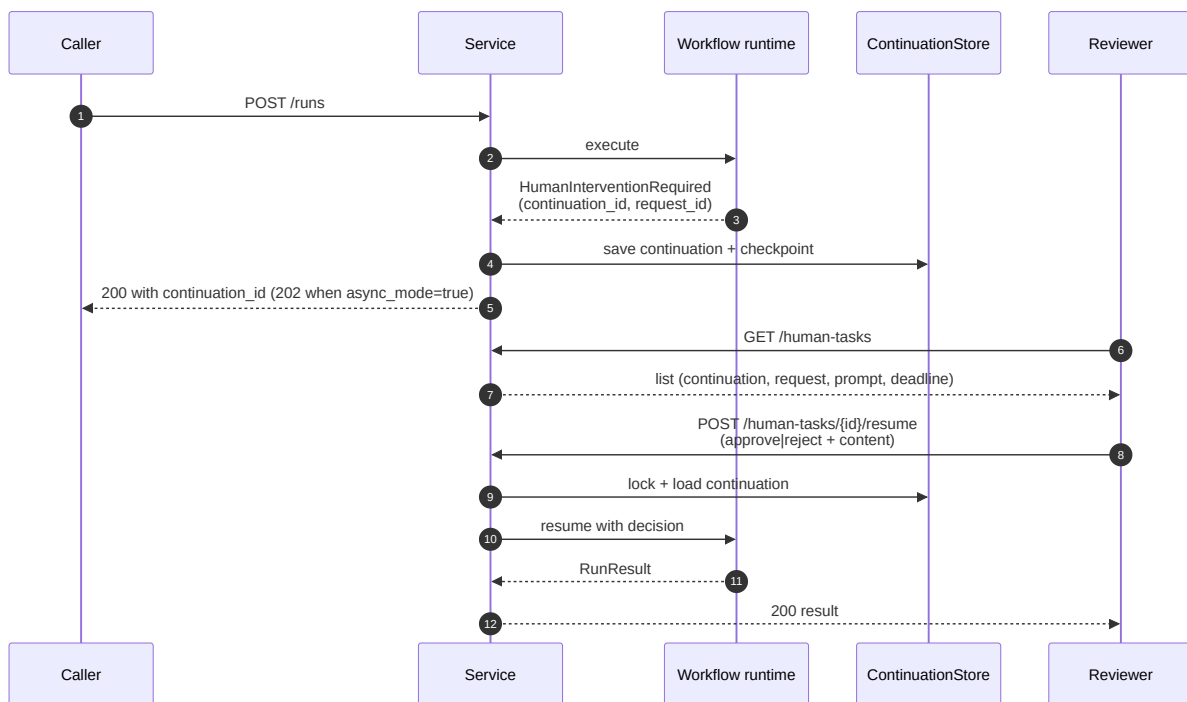
Source: docs/user/human_in_the_loop.md

An end-to-end guide to pause/resume workflows: declaring a human step in YAML, capturing the pause as a continuation, listing pending tasks, and resuming with a decision.

The architectural rationale lives in [design.md § 8.5](#); this page is the operational walkthrough.

What "human-in-the-loop" means here

A workflow can include steps that block on a human decision. When the runtime reaches such a step it raises `HumanInterventionRequired`, which the platform catches, serializes into a `WorkflowContinuation`, and exposes to operators. The run goes from `running` to `blocked`. A subsequent resume call provides the decision and the workflow continues from where it stopped, using the persisted replay context.



1 · Declare a human step in a spec

Human steps are supported in **sequential workflows only** — graph workflows have no pause/continuation path yet, so `/specs/validate` rejects a graph node with `kind: human` (`E_GRAPH_NODE_HUMAN_UNSUPPORTED`) rather than letting the run fail at the node.

[examples/human_approval_workflow.yaml](#) is the reference example. The relevant pieces:

```

workflow:
  type: sequential
  steps:
    - id: draft
      kind: function
      ref: draft_report
    - id: approve
      kind: human          # the pause point
      ref: approval-reviewer
    - id: publish
      kind: function
      ref: publish_report

components:
  humans:
    approval-reviewer:
      description: Please approve or edit the draft report.
      assignee: reviewer@example.com
      timeout_seconds: 86400
      on_timeout: escalate

```

A `human` step references an entry under `components.humans`, which provides the prompt, the assignee, and the timeout policy. The assignee is metadata for the operator UI; the platform doesn't dispatch notifications itself — wire that up at your routing layer.

Request taxonomy (optional)

Beyond `description` / `assignee` / `timeout_seconds` / `on_timeout`, a human component can declare what kind of decision it wants and how the reviewer's reply folds back in:

```

components:
  humans:
    pick-tier:
      description: Choose a support tier for this customer.
      request_type: selection          # approval | review | correction | selection | freeform
      options: [bronze, silver, gold] # choices a `selection` offers
      default_option: silver           # must be one of `options`
      context:                         # extra context surfaced to the reviewer
        sla_hours: 4
      response_role: assistant         # fold the reply into the thread as this role

```

All fields are optional and additive — `request_type` defaults to `approval` (the prior behavior). `validate_semantics` (and so `kneo spec validate`) rejects a `default_option` that is not one of `options`, and a `selection` with no `options`. When `response_role` is set, the reviewer's

provided/edited content is appended to the resumed run's message thread as a message with that role (otherwise it flows on only as the next step's input, as before).

2 · Run the workflow until pause

```
kneo run examples/human_approval_workflow.yaml \
  --input "draft the launch announcement" \
  --target workflow --json
```

When the workflow pauses on a human step, the response includes the run id, status, the `continuation_id` to resume against, and the pending request nested in `metadata.pending_human_request`:

```
{
  "run_id": "run_2026-05-10T12:34:56_...",
  "status": "blocked",
  "output_text": null,
  "human_intervention_required": true,
  "continuation_id": "cont_...",
  "metadata": {
    "pending_human_request": {
      "request_id": "req_...",
      "prompt": "Approve the draft?"
    }
  }
}
```

If you'd rather drive the API directly:

```
curl -sf -X POST http://127.0.0.1:8000/v1/runs \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" \
  -H 'Content-Type: application/json' \
  -d '{
    "input": "draft the launch announcement",
    "spec_path": "examples/human_approval_workflow.yaml",
    "target": "workflow"
  }'
```

The response carries the same `continuation_id` and `request_id` fields.

3 · List and inspect pending tasks

```
kneo human list --profile local
kneo human get <continuation_id>
```

Or via the API:

```
curl -sf "http://127.0.0.1:8000/v1/human-tasks?run_id=<run_id>" \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" | jq

curl -sf "http://127.0.0.1:8000/v1/human-tasks/<continuation_id>" \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" | jq
```

The single-task response carries the pending `request` and, when the paused run had one, the redacted `messages` thread leading up to the pause — so a reviewer UI can show the conversation in context, not just the prompt.

Listing requires the `human:read` scope; resuming requires `human:write`. See [production_readiness_review.md § Route Scope Matrix](#).

4 · Resume with a decision

```
kneo human resume <continuation_id> \
  --request-id <request_id> \
  --approve
```

Or with structured content over HTTP:

```
curl -sf -X POST "http://127.0.0.1:8000/v1/human-tasks/<continuation_id>/resume" \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" \
  -H 'Content-Type: application/json' \
  -d '{
    "request_id": "<request_id>",
    "decision": "approved",
    "content": "Looks good. Ship it."
  }'
```

`decision` must be one of `approved`, `rejected`, `edited`, `selected`, or `provided` (past-tense — the CLI's `--approve`, `--reject`, `--edit`, `--select`, and `--provide` flags map to these). `selected` pairs with a `selected_option`; `edited` and `provided` pair with `edited/provided content`. Which decisions a step accepts depends on its component definition; the full schema is in [service_api.md](#).

5 · Idempotent resume

`POST /human-tasks/{id}/resume` accepts an `Idempotency-Key` header. If you send the same key with the same body, the platform replays the original response instead of re-executing the resume. Mismatched bodies return `409 idempotency_key_conflict`.

```
curl -sf -X POST "http://127.0.0.1:8000/v1/human-tasks/<continuation_id>/resume" \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" \
  -H "Idempotency-Key: $(uuidgen)" \
  -H 'Content-Type: application/json' \
  -d '...'
```

In the CLI, `KNEO_SERV_IDEMPOTENCY_KEY` provides the header.

6 · Process-safe locking

The platform acquires a per-continuation lock before executing a resume. Two callers hitting `resume` on the same continuation see one succeed and the other receive `LockAcquisitionError`. This guarantees the same human task cannot be acted on twice.

If you see this error, wait for the in-flight resume to finish; do not retry blindly. See [troubleshooting.md § 8.1](#).

7 · Audit trail

Every human decision records an audit event:

```
curl -sf "http://127.0.0.1:8000/v1/audit-events?event_type=human.decision" \
  -H "Authorization: Bearer $KNEO_SERV_API_KEY" | jq
```

The payload records `request_id`, decision, selected option, result status, and *whether* content was present — never the content itself. See the audit policy in [production_readiness_review.md § Audit Payload Review](#).

8 · Recovery after restart

`WorkflowContinuation` is persisted in `ContinuationStore`. If the service restarts mid-run, the paused continuation survives. After restart:

- Listing `/human-tasks` returns the same continuation.
- Resume picks up from the persisted replay context — the workflow does not retry completed steps.

If a non-human workflow is interrupted (e.g., a worker crash), the same mechanism enables continuation. Check `/runs/{id}/recovery` to see whether continuation is available, then call `/runs/{id}/continue`.

9 · Timeouts and escalation

`components.humans.<id>.timeout_seconds` and `on_timeout` declare the policy. When a sequential workflow pauses on a human step that declares a positive `timeout_seconds`, the platform computes `expires_at = pause_time + timeout_seconds` and stashes it on the continuation along with the chosen `on_timeout` value. The prune step below dispatches on that value (`fail` / `continue` / `escalate`). Auto-routing of an escalated task to a different reviewer is still up to the operator's external workflow — the platform marks the task as escalated and emits an audit event, but does not auto-reassign.

Expiring paused runs

Call `PlatformManager.prune_expired_human_tasks()` on whatever cadence the deployment needs (cron, scheduled run, manual operator action — the same pattern as `prune_retention()`; there is no built-in scheduler).

For each saved continuation whose `expires_at` is in the past and whose underlying run is still `blocked`, the prune dispatches on `pending_human_request["on_timeout"]`:

- **fail** (default): marks the run `expired` (a lifecycle status alongside `failed` / `cancelled`); `run.error.type` is `human_task_expired` with a message recording the configured timeout. Records a `human.expired` audit event with `run_id`, `continuation_id`, `timeout_seconds`, `expires_at`, `expired_at` (the cutoff used), and `on_timeout`. Deletes the continuation.
- **continue**: synthesizes an `approved` `HumanResponse` carrying `metadata.auto_continued = true`, `metadata.reason = "timeout"`, and the original `metadata.original_assignee`, then resumes the workflow past the paused step. Records a `human.continued` audit event before the resume attempt; if the resume itself raises, also records `human.continue_failed` (with `error_type` and `error_message`) and deletes the continuation so the prune does not retry indefinitely.
- **escalate**: keeps the run `blocked` and stamps `pending_human_request["escalated_at"]` (plus a copy of the original `expires_at` as `original_expires_at`) on the continuation. Records a `human.escalated` audit event including the original `assignee`. Subsequent prune calls **skip** continuations carrying the `escalated_at` marker — escalation fires once. The continuation stays alive until an operator resumes it via the normal `/v1/human-tasks/{continuation_id}/resume` path (typically after reassigning to a different reviewer in the operator's external workflow).

Runs that have already resumed before the prune fires (status no longer `blocked`) are skipped on every branch — the resume path owns the terminal state. Calls are idempotent on the `fail` and `escalate` paths; the `continue` path is single-shot per continuation by construction (success deletes via the resume; failure deletes explicitly).

`prune_expired_human_tasks()` returns the count of continuations processed this call across all branches. Already-escalated continuations are skipped and do not contribute to the count.

Run-level timeouts vs. human-task timeouts

Two independent timeouts can apply to a run that's blocked on a human step:

- **Human-task timeout** (`components.humans.<id>.timeout_seconds + on_timeout`): bounded waiting time on *this specific human step*. Stored on the continuation as `expires_at`. Handled by `prune_expired_human_tasks()` per the dispatch above.
- **Run-level timeout** (`start_run_from_spec(..., timeout_seconds=N)`): bounded wall-clock for the *whole run*, no matter which step it's currently on. Stored on the run state as `deadline_at`. Handled by `PlatformManager.prune_timed_out_runs()`.

Whichever fires first wins. If the run-level prune fires while the run is blocked on a human task, the run transitions to `timed_out` (not `expired`), the continuation is deleted, and a `run.timed_out` audit event is recorded — the human-task path is preempted. If the human-task prune fires first, the human-task `on_timeout` semantics apply as documented above; the run-level deadline becomes irrelevant because the run is no longer in `blocked` status.

The `timed_out` lifecycle status is distinct from `expired`: `expired` means a human task missed its deadline (the run was waiting on a human); `timed_out` means the run as a whole missed its deadline (could have been waiting on a human, could have been mid-execution).

Common failure modes

Symptom	See
<code>LockAcquisitionError</code> on resume	troubleshooting.md § 8.1
404 on continuation_id	troubleshooting.md § 8.2
409 <code>idempotency_key_conflict</code>	troubleshooting.md § 5.4
403 Missing required scope: <code>human:write</code>	troubleshooting.md § 4.2

See also

- [examples.md](#) — the full set of runnable specs.
- [service_api.md](#) — `/human-tasks` route shapes.
- [design.md § 8.5](#) — design rationale.